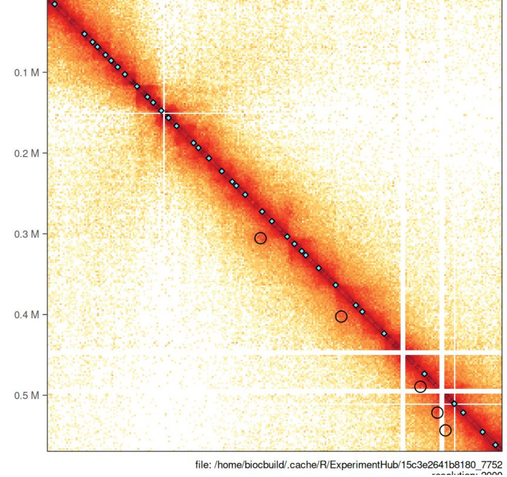
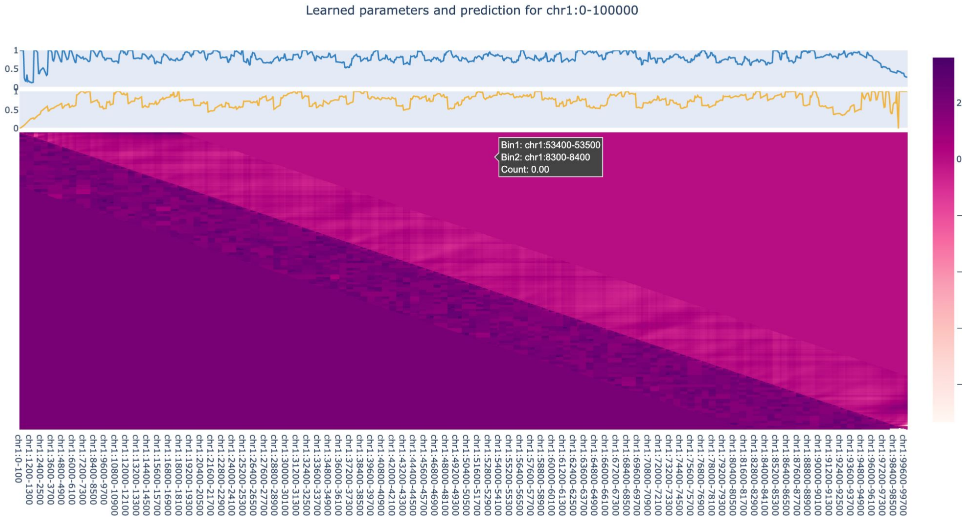
Research update

Henry Kuerbis
3/11/26

Hi-C contact matrices and dLEM

- Hi-C sequencing measures the frequency of contact between specific genomic loci
- The result is a heatmap
- 2D heatmap (Hi-C) → 1D cohesion movement parameters → 2D heatmap (dLEM)





Evaluating Robustness of dLEM Parameters on Noisy Data

- How much information is stored in and recovered from the parameters fit by dLEM?
- How does noise in the input data affect the models ability to make accurate predictions?

Basic workflow

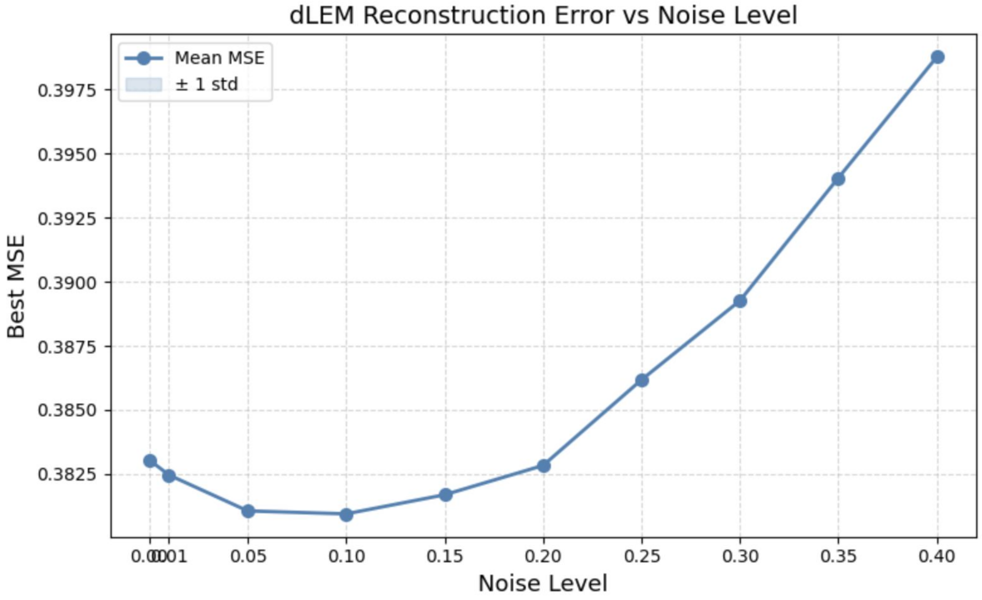
1. Generate synthetic Hi-C contact matrices
2. Inject controlled levels of noise
3. Run dLEM on noisy matrices
4. Measure model performance
5. Evaluate robustness of matrix and parameter recovery

Noise generation and initial results

Noise is added to each pixel in the matrix:

$M_{ij}^{*} = M_{ij} + \text{rand}(-1,1) \times \text{noise\_level} \times M_{ij}$

- MSE not great
- Some noise improves model prediction
- Would like to compare noisy predictions to original non noisy matrix, to see how well signal is captured through noise.



Improvements in synthetic data

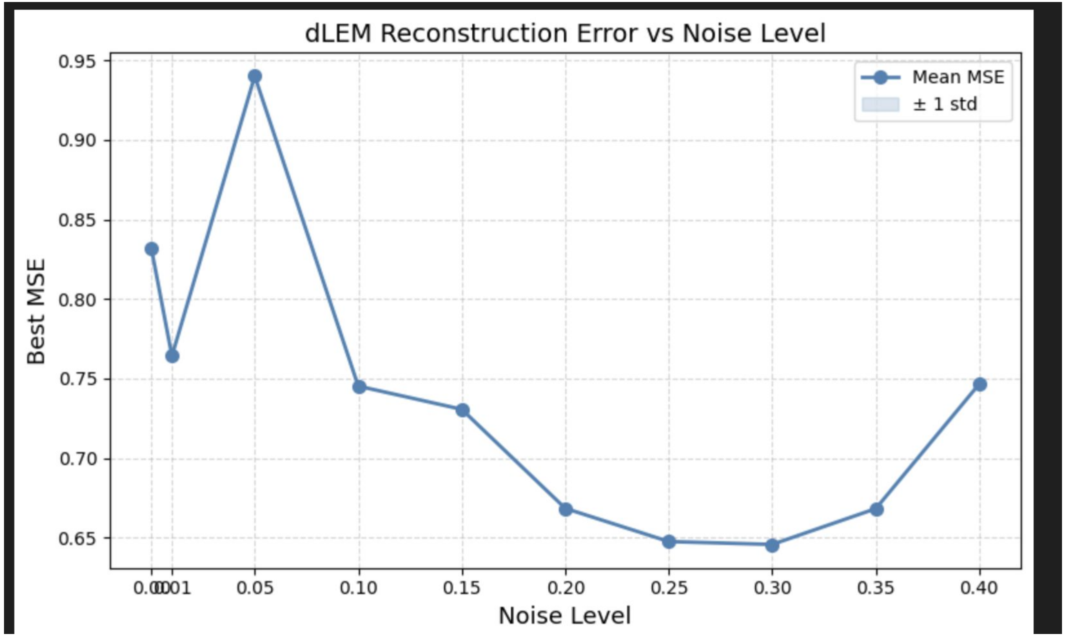
Need “theoretically perfect data” to make very accurate starting prediction

- Then the effect of noise on model prediction will be better isolated

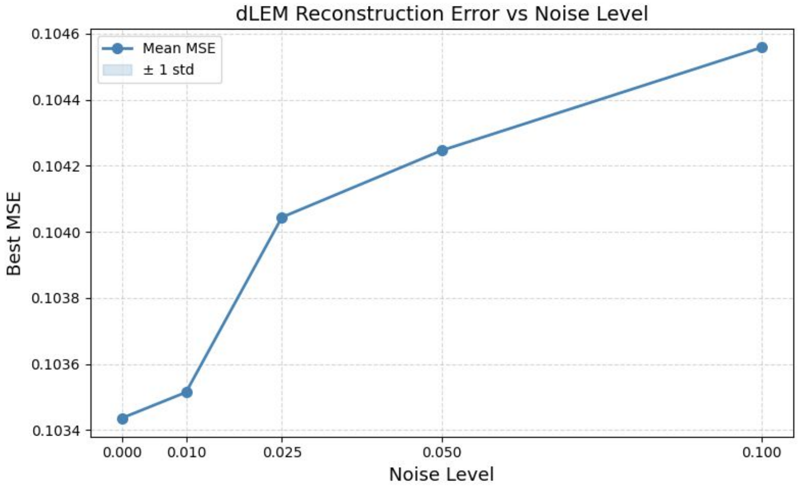
Currently the synthetic matrix is being built by Diego to resemble experimental data.

Next step will be to use dLEM to build a predicted matrix from Diego's synthetic matrix, to hopeful get a stronger starting prediction. Many iterations?

Maybe look existing work on cohesion dynamics simulation.


dLEM Reconstruction Error vs Noise Level
Mean MSE
± 1 std
Best MSE
Noise Level
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.00
0.01
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40

chr3R



dLEM Reconstruction Error vs Noise Level
Mean MSE
± 1 std
Best MSE
Noise Level
0.1046
0.1044
0.1042
0.1040
0.1038
0.1036
0.1034
0.000
0.010
0.025
0.050
0.100